

☐ Differentiation of binary cross entropy loss function

Here,

Binary cross entropy loss -

$$L = -[y \log(a) + (1-y) \log(1-a)]$$

$$\therefore dL/da = - d/da [y \log(a) + (1-y) \log(1-a)]$$

$$\therefore d/da [y \log(a)] = y/a \text{ And,}$$

$$d/da [(1-y) \log(1-a)] = - \frac{1-y}{1-a}$$

$$\therefore dL/da = - \left(y/a - \frac{1-y}{1-a} \right) = -y/a + \frac{1-y}{1-a}$$
$$= \frac{-y(1-a) + a(1-y)}{(1-a) \cdot a}$$

Here,

$a = \sigma(z)$ (sigmoid output) -

$$da/dz = a(1-a)$$

According to chain rule -

$$dL/dz = dL/da \times \frac{da}{dz} = \frac{a-y}{a(1-a)} \times a(1-a)$$
$$= a-y$$

Ans!